Layout experiment 1.a: before and after session 2.b

Companion to the design review (Sofar's mote beside this board, region by region).

Before = the board session 2 ended with (commit b22752f, 46 × 65 mm). After = the board now (commit b34293d, md5 23e5e8f9…, 49 × 68 mm). Same renderer for both (`tools/render_layers.py`), same review regions (in each board's own frame). Generated by `tools/before_after.py` from `tools/ba_data.py` and `tools/summary.py`.

The short version

The board grew and the ADIN moved to the middle. 138 footprints before, 138 after; **79** changed position, rotation or side (the whole ADIN block, both transformer clusters, the 1.8 V buck, the port-1 side with L1, the inserts, the jumpers, the fiducials and the south band). The frame's origin (the Pi's north-west corner) is the same in both.

Measure	Before (b22752f)
Board	46 × 65
SPI copper, new track mm (SCK / MOSI / MISO / ~CS)	— / — / 66.5 / 67.6 (west-edge detour)
Pairs (mm; limits 9.5 / 9.5 / 21.3 / 21.3)	9.46 / 8.80 / 17.98 / 18.62
Segments per pair leg (the staircases, QE N3)	25 / 40 per leg (staircases)
Open links besides J1's no-connect pair	0
DRC	111 errors (all Sofar's), 254 warnings, 1 unconnected
Rules: below class / thin links / clearance items	0 / 0 / 0
JLCPCB DFM table	22 pass / 9 fail (DFM.md History)
Vias in all	277

Copper per layer

Layer	Before: segments / mm	After: segments / mm
Top Layer	444 / 301.2	381 / 321.5
Internal 1	311 / 183.8	134 / 194.0
Internal 2	862 / 442.9	186 / 431.5
6 Bottom Layer	521 / 260.4	504 / 268.3
vias	277	265
filled zones	15	15

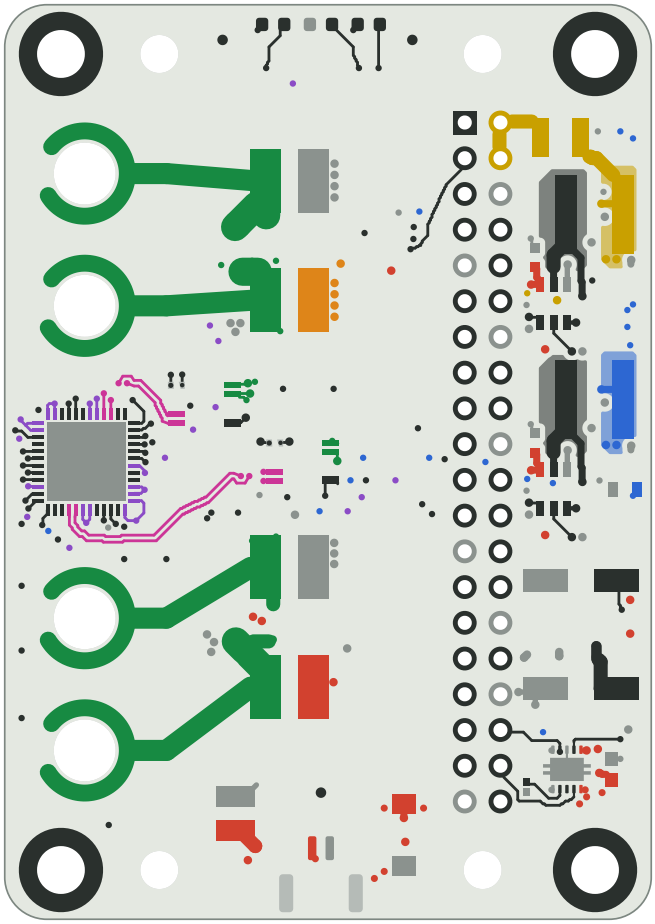
Nets whose copper changed by 2 mm or more, or whose via count changed

Net	Before mm	After mm	Vias	Layers before → after
1V8	36.7	34.8	4 → 3	Bottom, Internal 1, Internal 2 → Bottom, Internal 1
3V3	111.4	113.8	21 → 16	Bottom, Internal 1, Internal 2, Top → Bottom, Inter
5V_PI	9.1	22.2	5 → 8	Bottom, Internal 1, Top → Bottom, Internal 1, Top
ADIN_LED_VDD	6.6	5.6	0 → 2	Bottom → Bottom, Top
ADIN_MISO	65.5	38.7	1 → 3	Internal 1, Internal 2, Top → Internal 1, Internal 2,
ADIN_MOSI	32.9	33.0	2 → 1	Internal 1, Internal 2, Top → Internal 1, Internal 2,
ADIN_PWR	42.6	42.3	10 → 12	Bottom, Internal 1, Internal 2 → Bottom, Internal 1
ADIN_SCK	32.2	32.2	3 → 1	Internal 1, Internal 2, Top → Internal 1, Internal 2,
ADIN_VDDIO	47.4	50.8	10 → 6	Bottom, Internal 1, Internal 2, Top → Bottom, Inter
BM1_N	57.6	65.5	12 → 12	Bottom, Internal 1, Internal 2, Top → Bottom, Inter
BM1_P	42.8	49.6	11 → 10	Bottom, Internal 1, Internal 2, Top → Bottom, Inter
BM2_DATA_N	18.2	20.9	2 → 2	Internal 1, Top → Internal 1, Top
BM2_DATA_P	17.5	20.1	2 → 2	Internal 1, Top → Internal 1, Top
BM2_N	38.1	46.7	12 → 10	Bottom, Internal 1, Top → Bottom, Internal 1, Top
BM2_P	51.0	59.5	12 → 12	Bottom, Internal 1, Internal 2, Top → Bottom, Inter
GND	53.8	57.3	71 → 71	Bottom, Top → Bottom, Top
Net-(U4-ALERT)	5.2	8.4	2 → 2	Bottom, Internal 1 → Bottom, Internal 1, Internal 2
PAYLOAD_EN	15.6	17.8	1 → 1	Bottom, Top → Bottom, Top
PI_5V	2.3	11.2	0 → 1	Top → Internal 2, Top
VBUS	28.5	31.0	18 → 18	Bottom, Top → Bottom, Top
VBUS_OUT	23.1	23.1	9 → 7	Bottom, Internal 2, Top → Bottom, Internal 2, Top
~{ADIN_CS}	65.2	40.3	6 → 2	Internal 1, Internal 2, Top → Internal 1, Internal 2,
~{ADIN_INT}	39.7	42.4	1 → 2	Bottom, Internal 2, Top → Bottom, Internal 2, Top
~{ADIN_P1_LED1}	33.4	37.8	4 → 3	Bottom, Internal 1, Internal 2, Top → Bottom, Inter

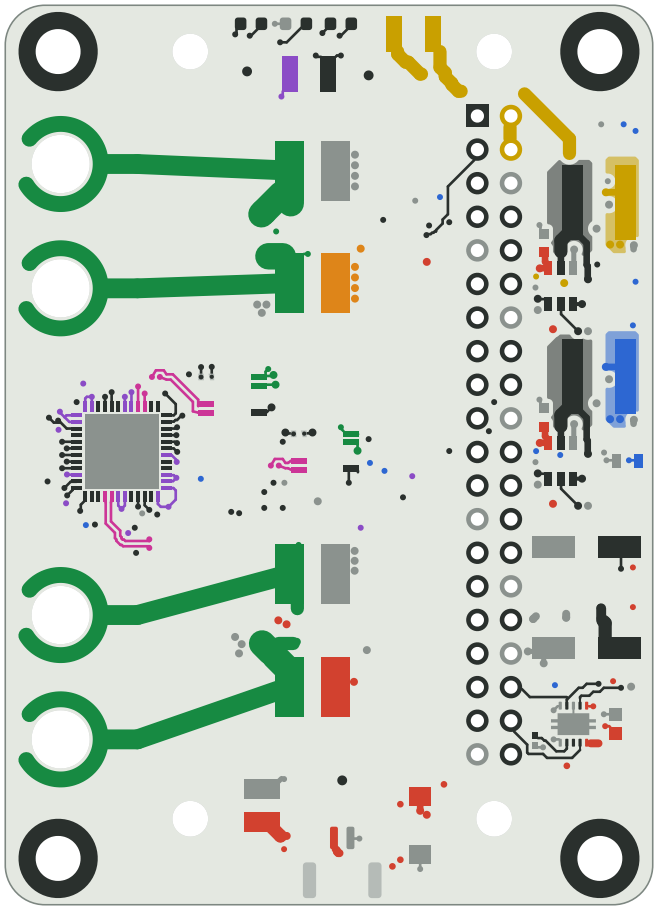
Whole board, layer by layer

Left before (46 × 65), right after (49 × 68); each in its own frame. Colours: magenta data pairs, green bus, red VBUS, blue 3V3, purple rails, yellow 5 V to the Pi, dark signals, grey GND fills.

Top

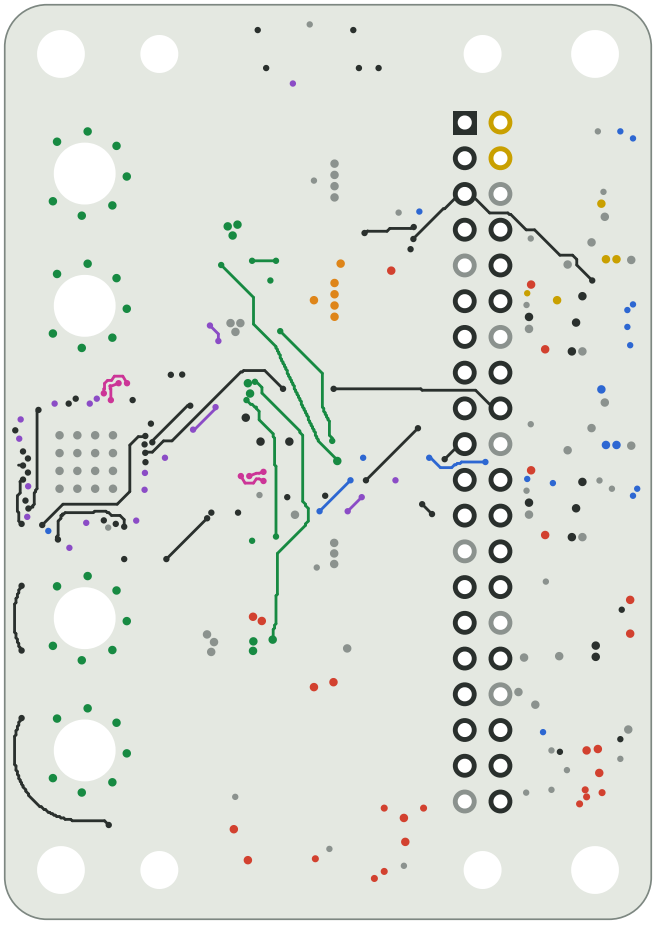


Before, Top

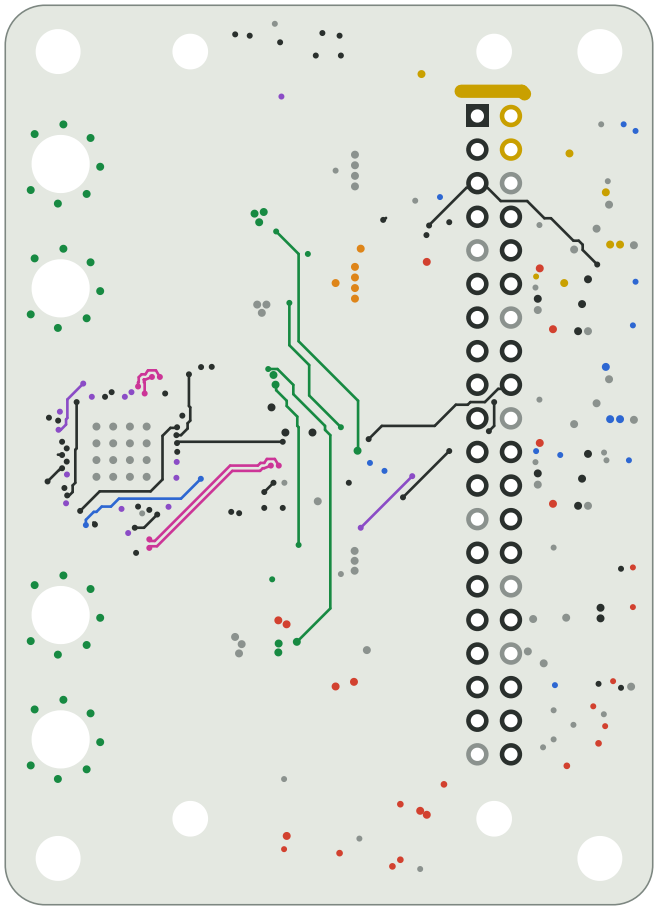


After, Top

Internal 1

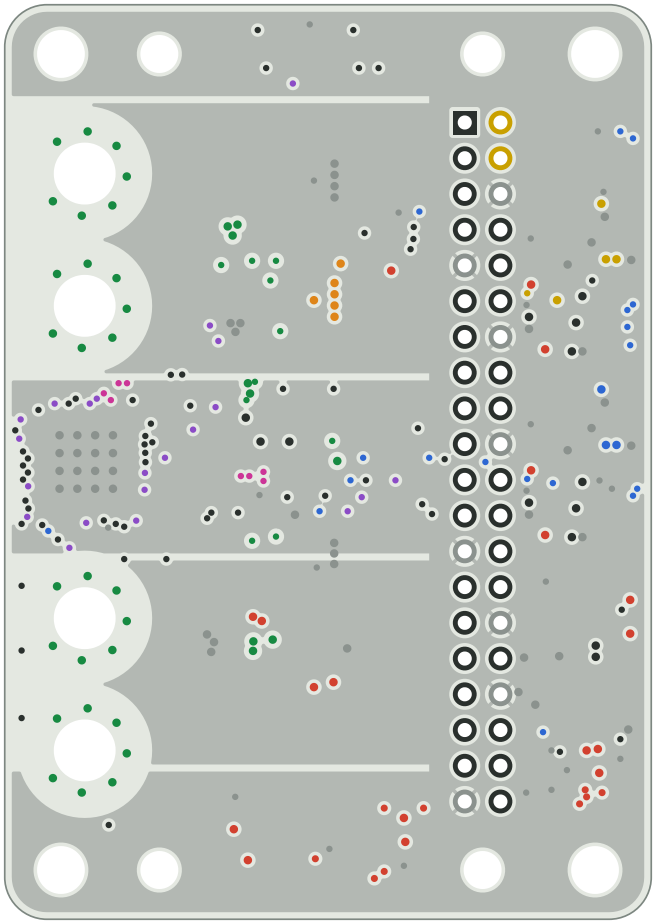


Before, Internal 1

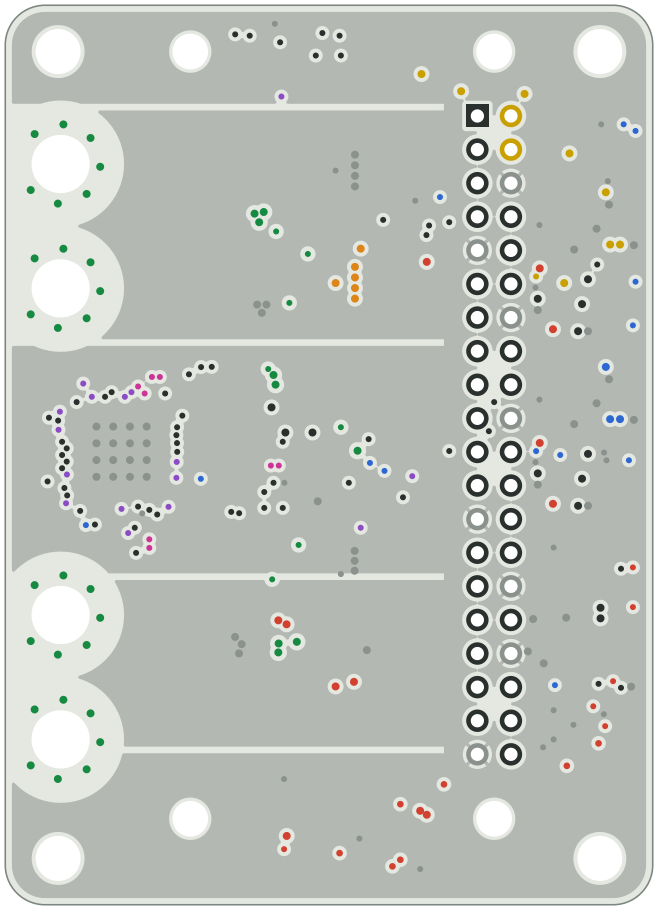


After, Internal 1

GND plane (In2)

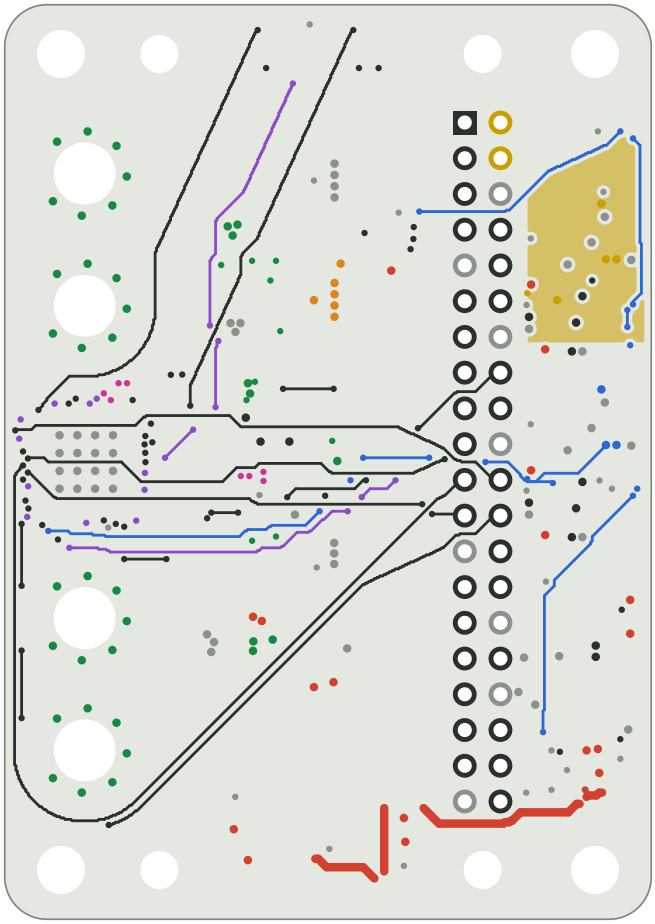


Before, GND plane (In2)

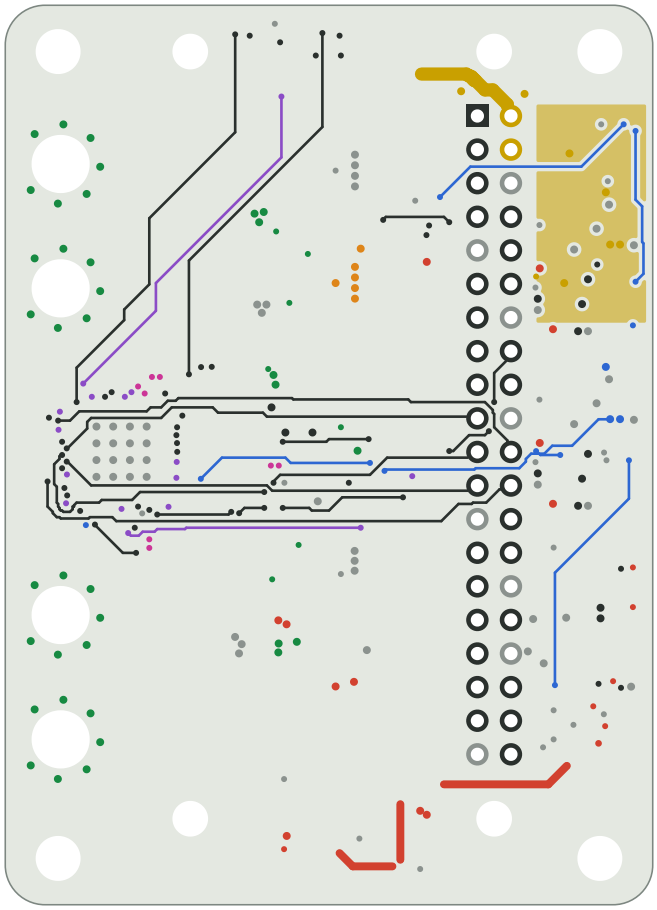


After, GND plane (In2)

Internal 2 (In3)

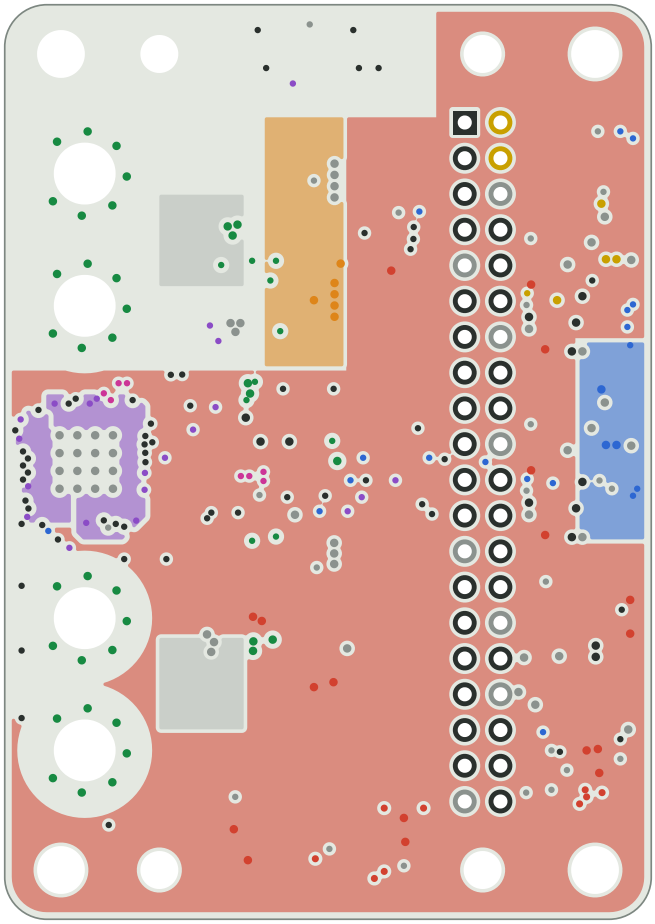


Before, Internal 2 (In3)

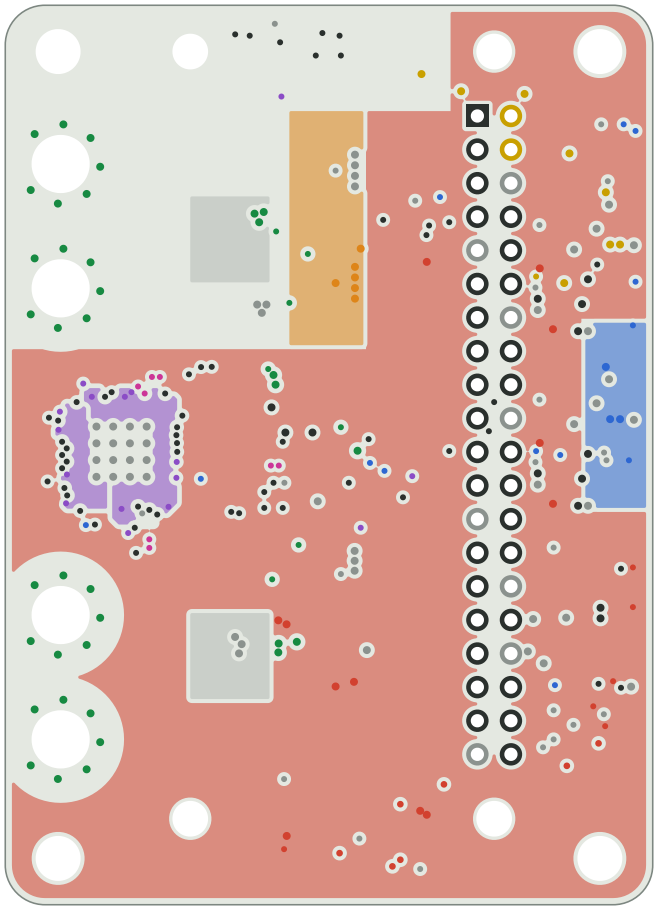


After, Internal 2 (In3)

PWR plane (In4)

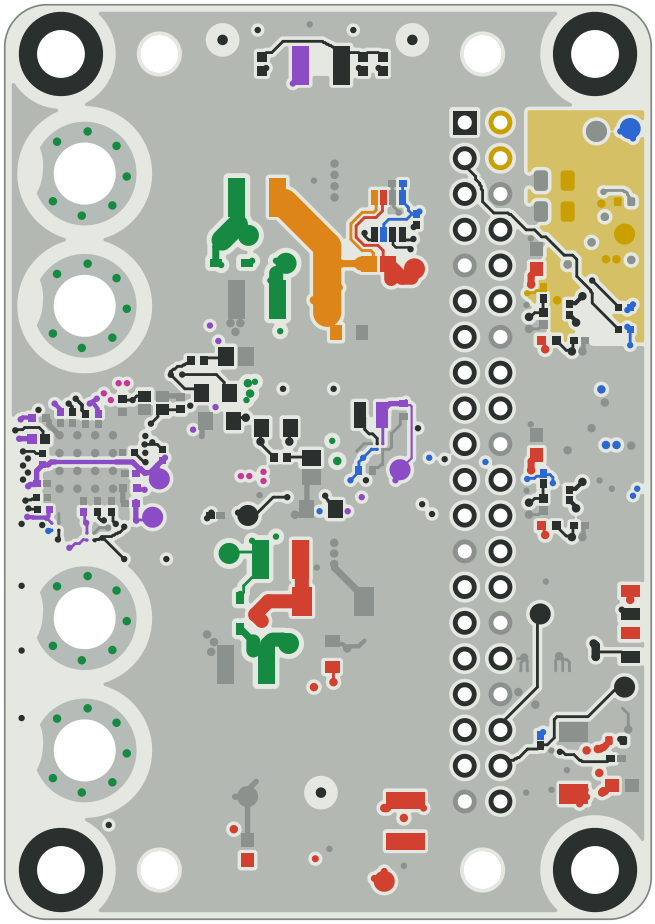


Before, PWR plane (In4)

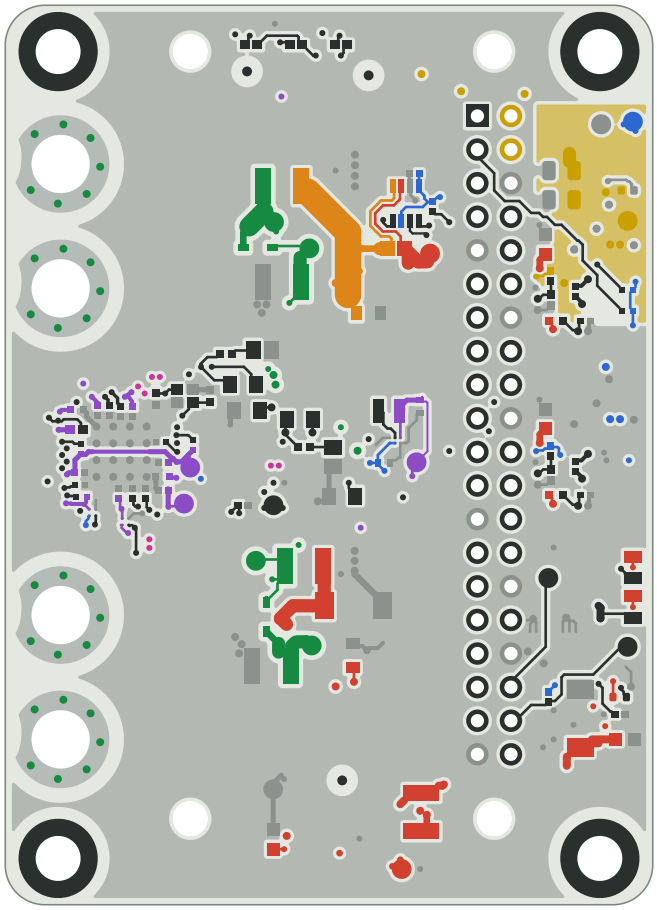


After, PWR plane (In4)

Bottom



Before, Bottom



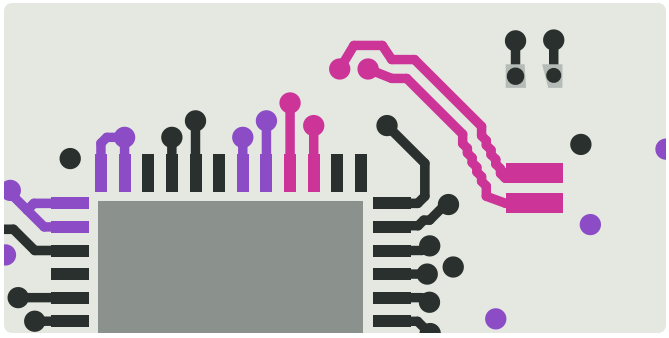
After, Bottom

The review regions, before and after

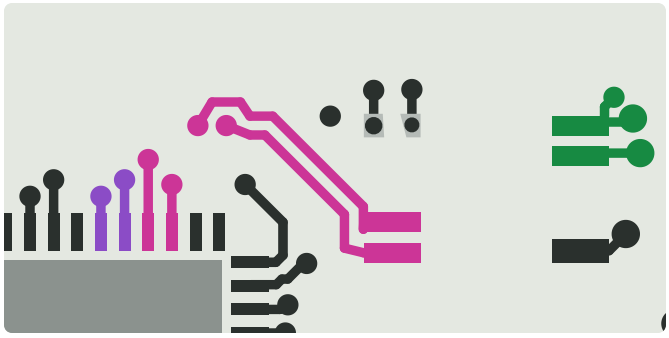
The regions are those of the design review (the new placement); on the 'before' board the same rectangle shows whatever sat there.

port1 pair x -3.5…10.5, y 25.5…32.5 mm

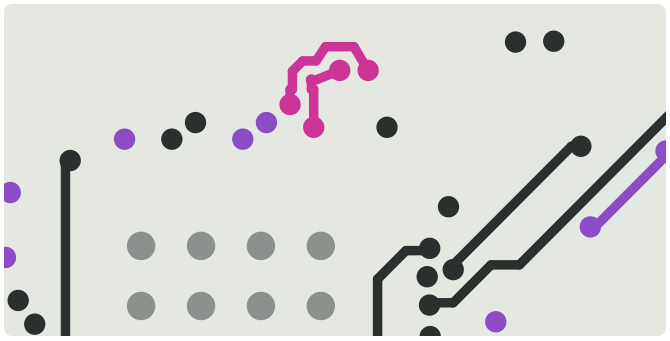
Port-1 data pair: U1 pins 27/28 (north edge of U1 at rot 90) to T1 pads 1/2 in the band; the mote's U1 and T1 sit elsewhere, so the mote view shows only the ADIN block's copper placed as here



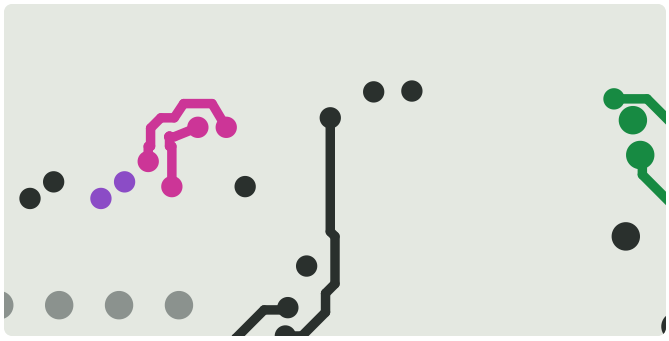
Before, Top



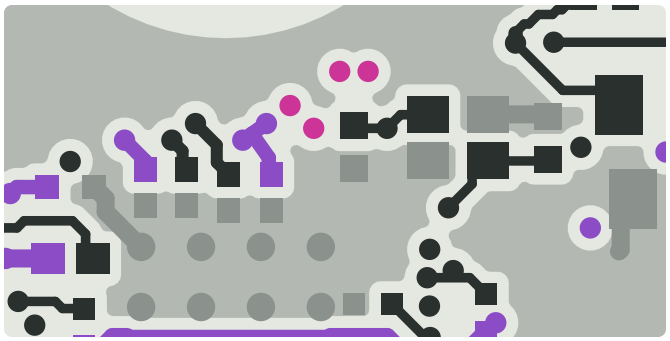
After, Top



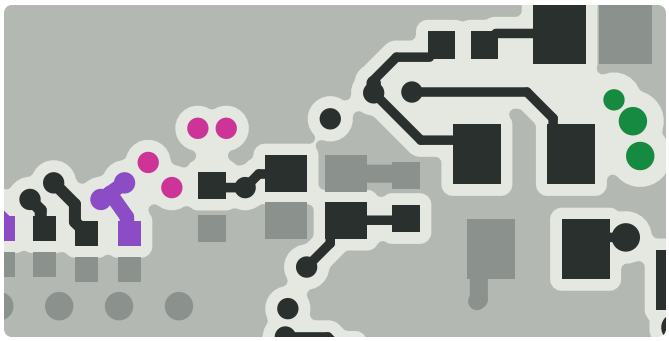
Before, Internal 1



After, Internal 1



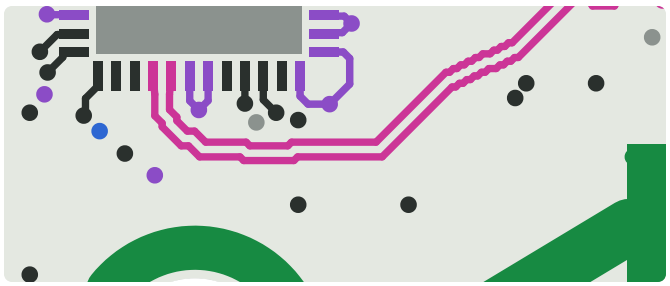
Before, Bottom



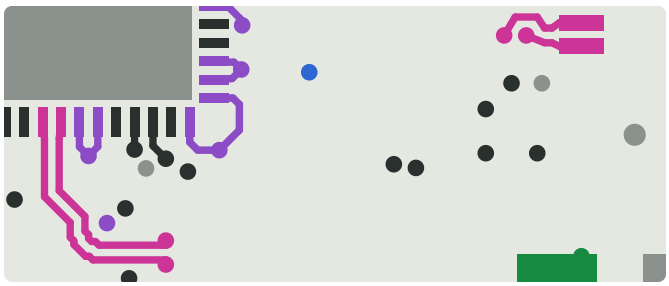
After, Bottom

port2 pair x -4.0…14.0, y 34.0…41.5 mm

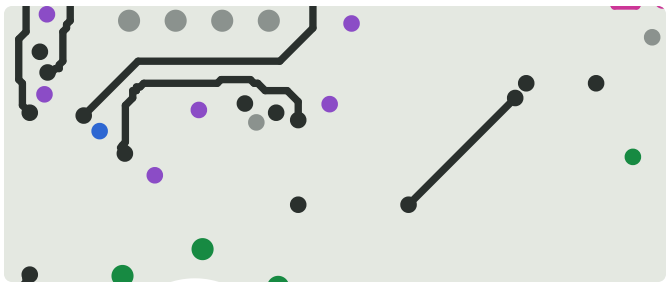
Port-2 data pair: U1 pins 4/5 (south edge) through the pocket's south exit to T2 pads 1/2



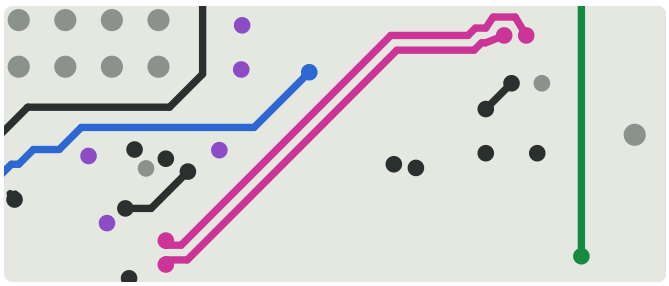
Before, Top



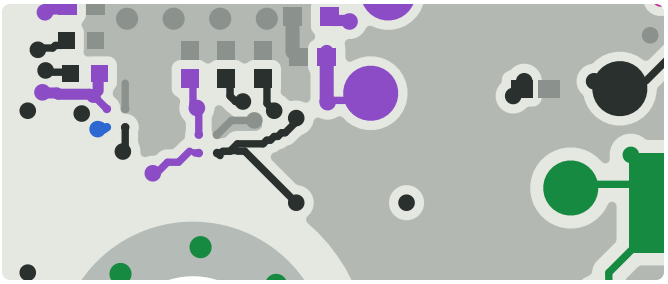
After, Top



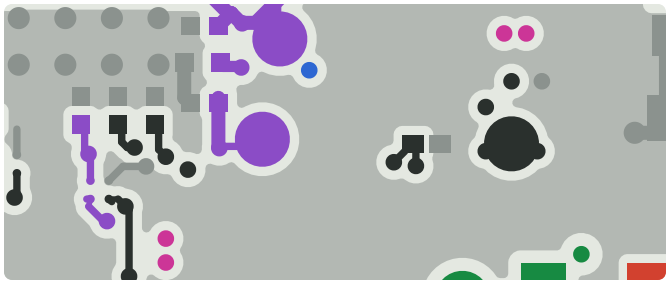
Before, Internal 1



After, Internal 1



Before, Bottom



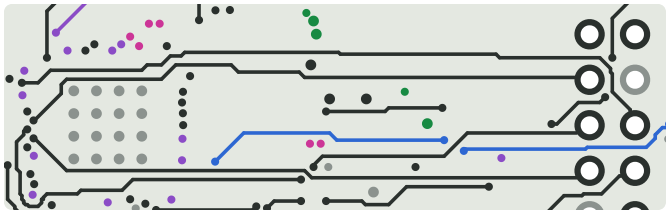
After, Bottom

spi lane x -7.5…29.5, y 27.0…38.5 mm

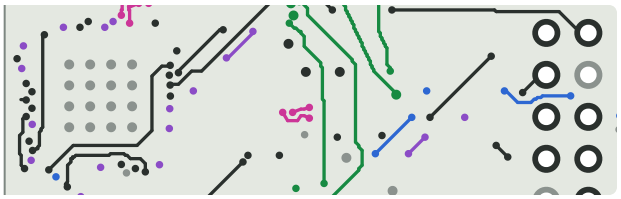
The SPI lane: from Sofar's vias at U1's west side east on Internal 2 under U1 and across the band to J1 pins 19-24, routed first (session 2: 25 mm down the west edge past the inserts)



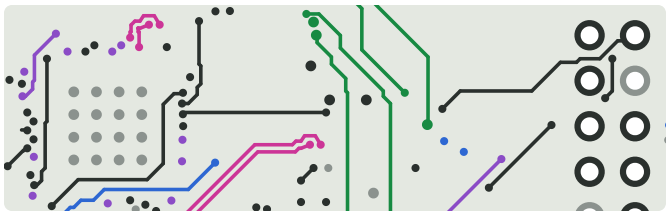
Before, Internal 2 (In3)



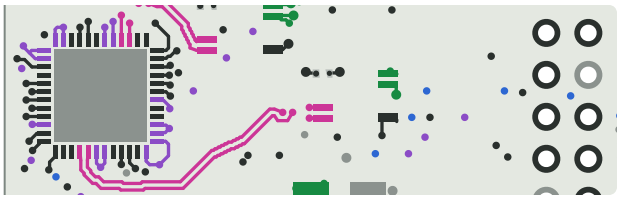
After, Internal 2 (In3)



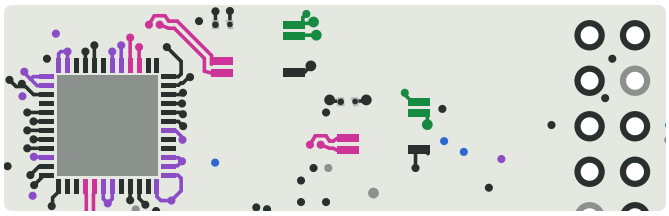
Before, Internal 1



After, Internal 1



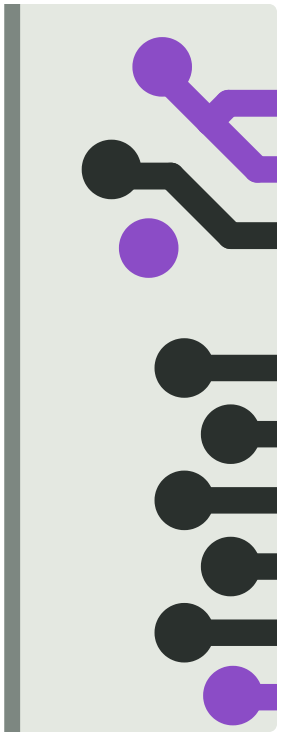
Before, Top



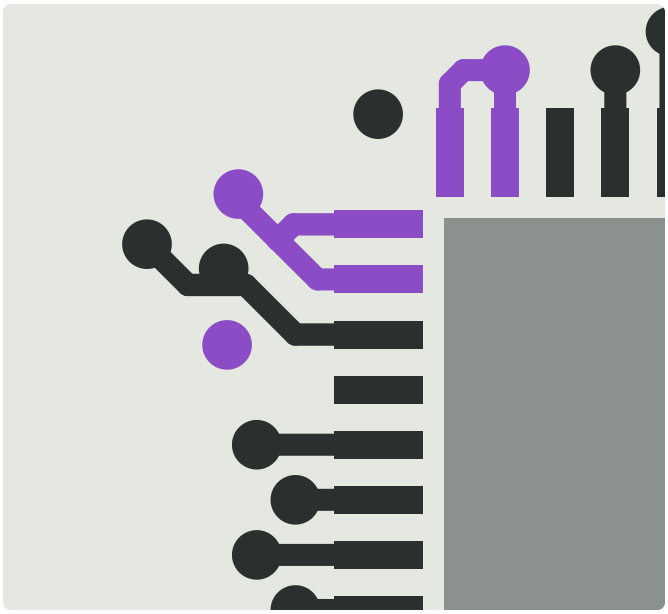
After, Top

int fanout x -8.5…-2.5, y 29.0…34.5 mm

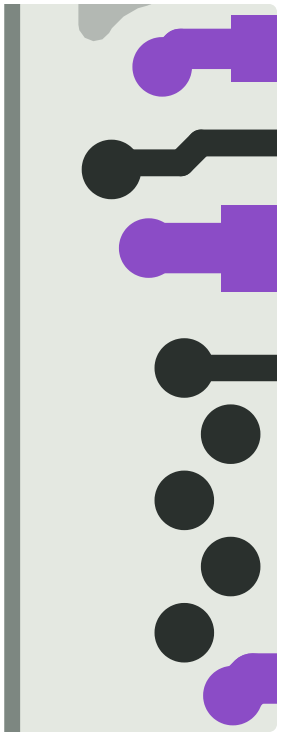
~{ADIN_INT} fan-out west of U1: U1.39 -> Sofar's own via (2.8 mm inside the edge now that U1 is 3 mm east: no hand via) -> R1.1



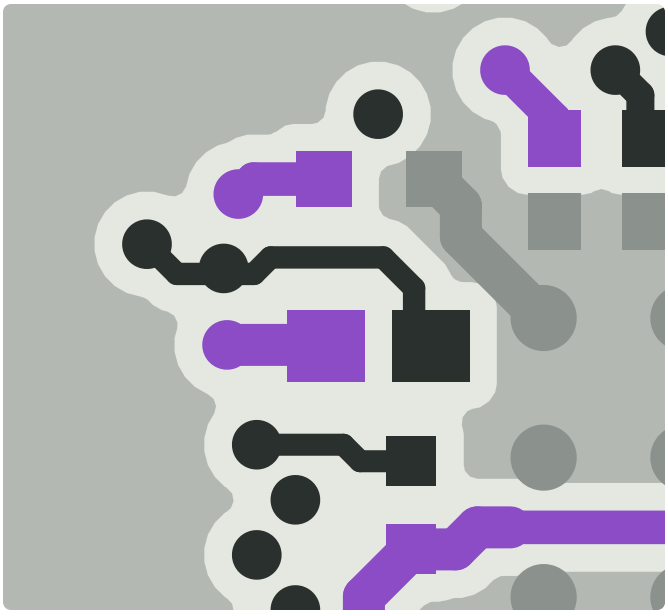
Before, Top



After, Top



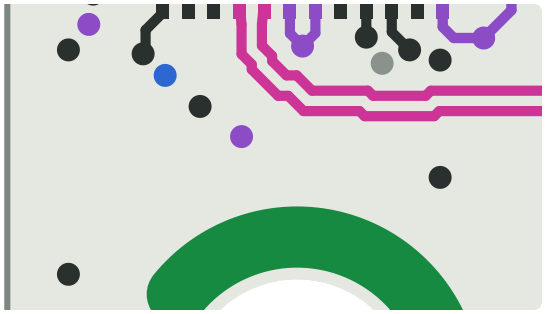
Before, Bottom



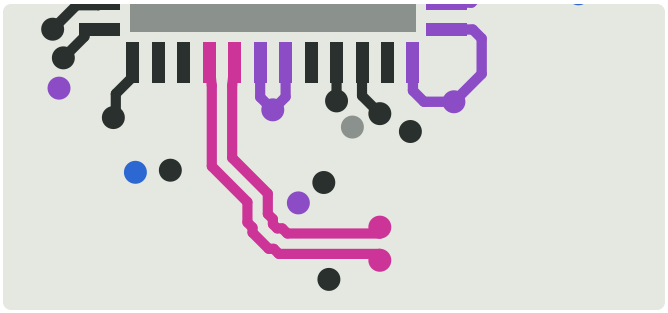
After, Bottom

u2 u3 pocket exit x -7.0…6.0, y 36.0…42.0 mm

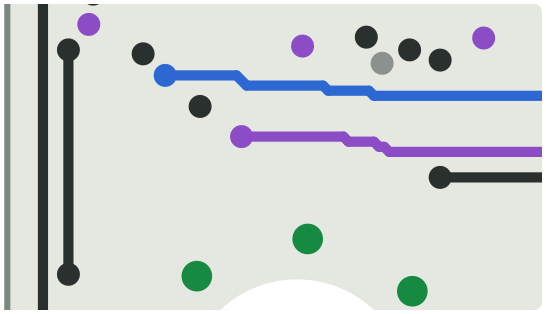
The pocket's south exit: U2 / U3 (1.8 V and 3.3 V switches, bottom) with their fan-out vias, the port-2 pair above, 3V3 / 1V8 / ADIN_PWR leaving east; 3 mm taller than session 2



Before, Top



After, Top



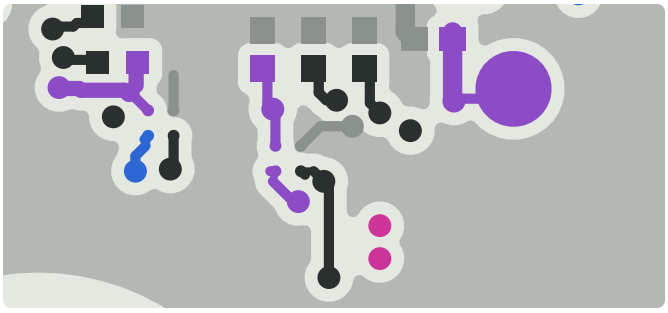
Before, Internal 2 (In3)



After, Internal 2 (In3)



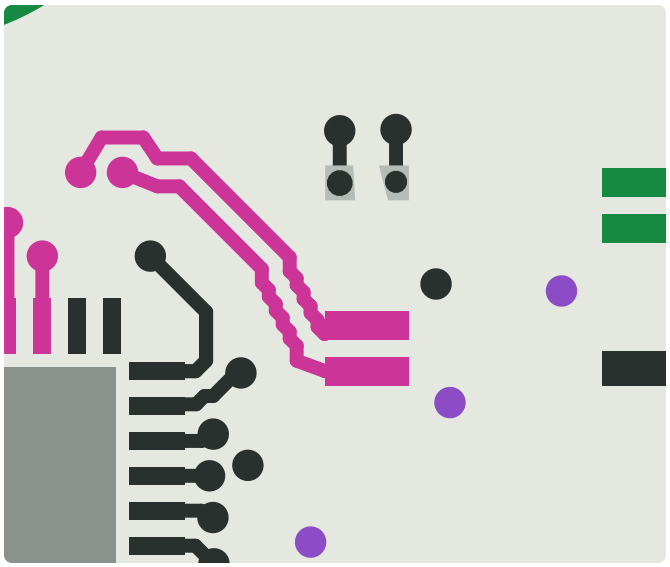
Before, Bottom

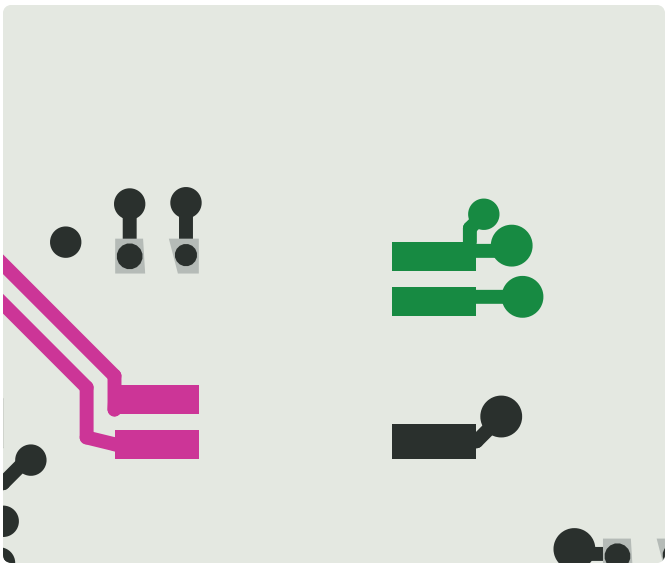


After, Bottom

t1 cluster x 2.5…12.0, y 24.5…32.5 mm

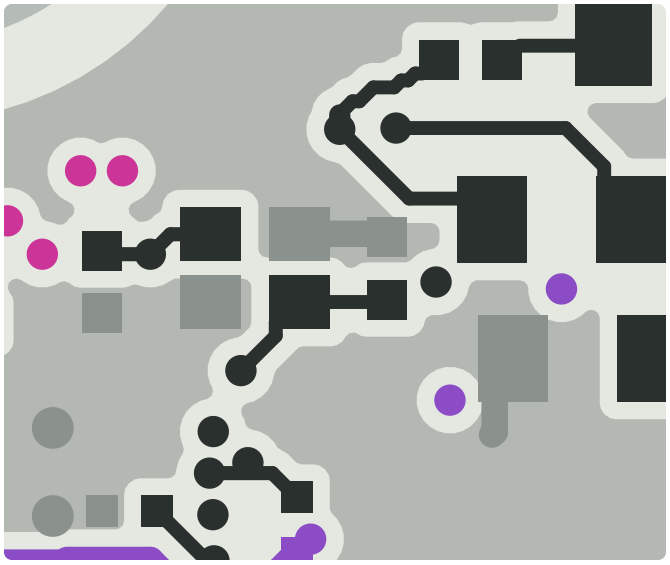
T1 cluster (P1T block) and the bus legs to L1



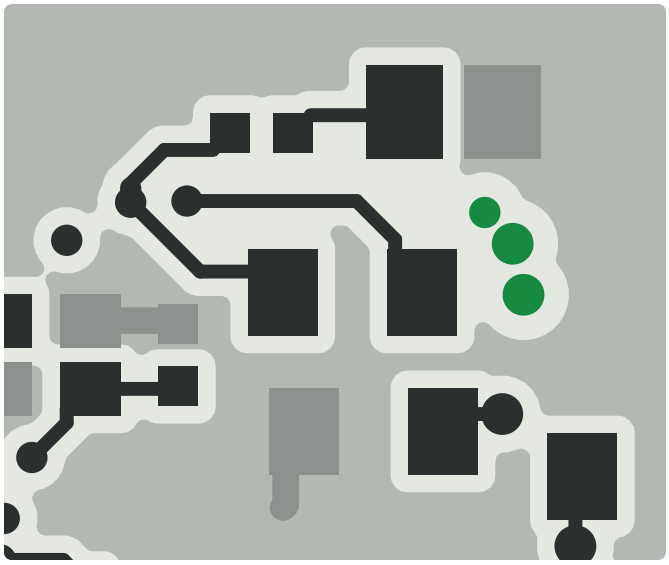


Before, Top

After, Top



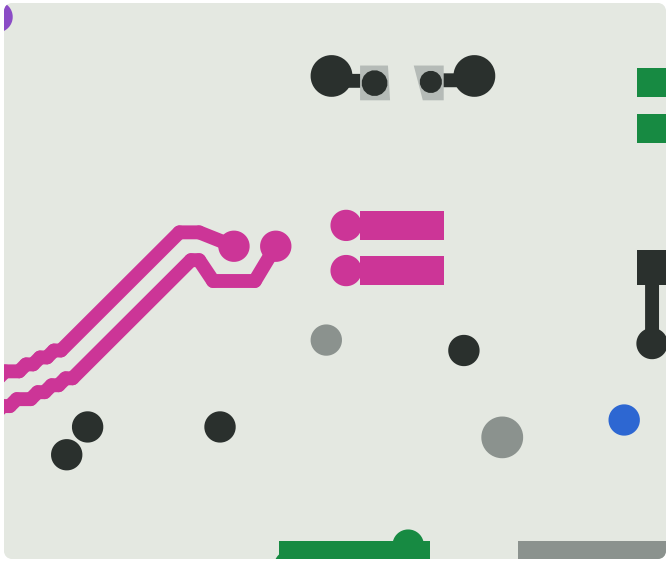
Before, Bottom



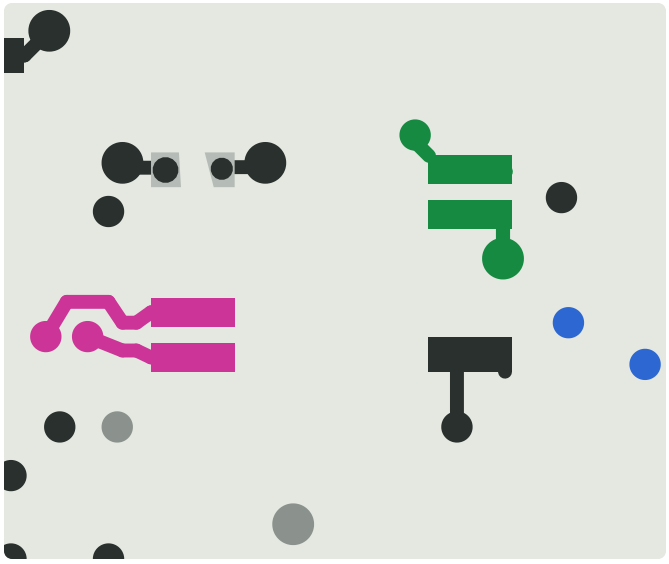
After, Bottom

t2 cluster x 9.0…18.5, y 30.0…38.0 mm

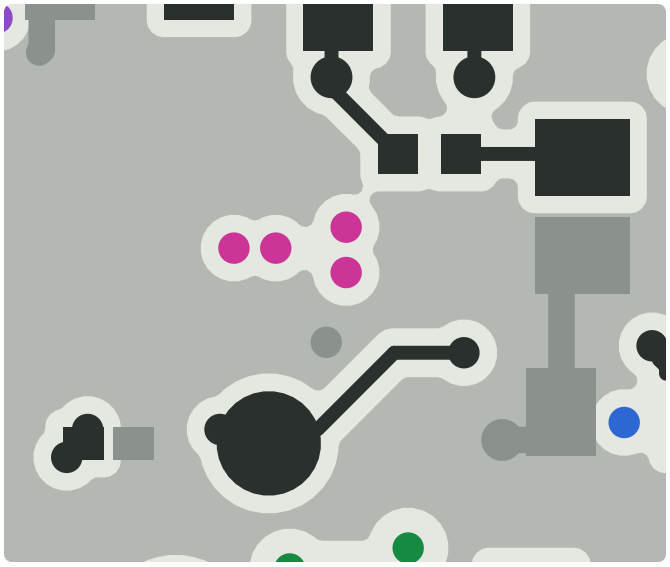
T2 cluster (P2T block) east of T1



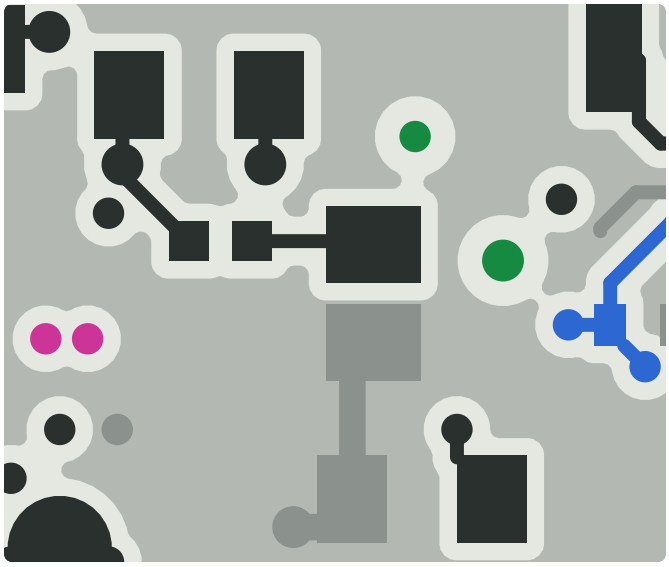
Before, Top



After, Top



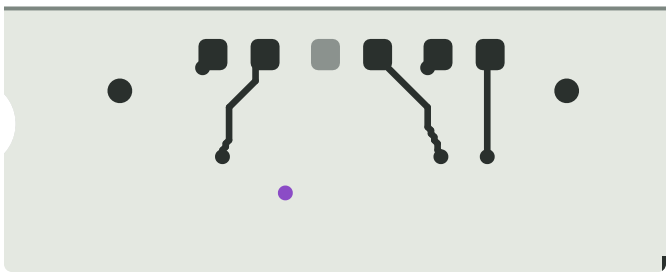
Before, Bottom



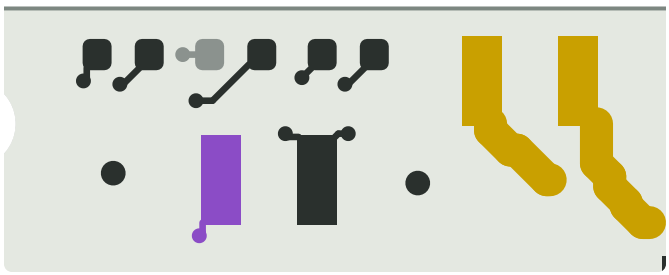
After, Bottom

north band x 4.5…24.5, y -0.5…8.0 mm

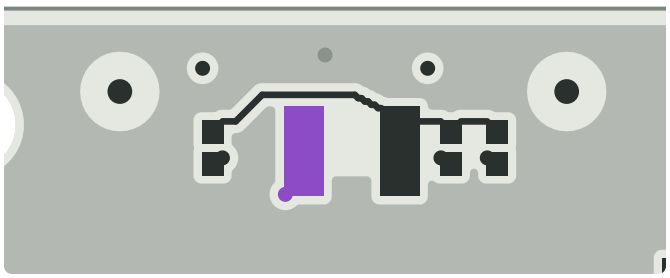
North band: the LEDs D10 / D8 / D9 on the edge, JP1 (5 V to the Pi) and JP2 (LED supply) on top beside them, FID3 / FID4; the LED resistors and FID1 / FID2 underneath



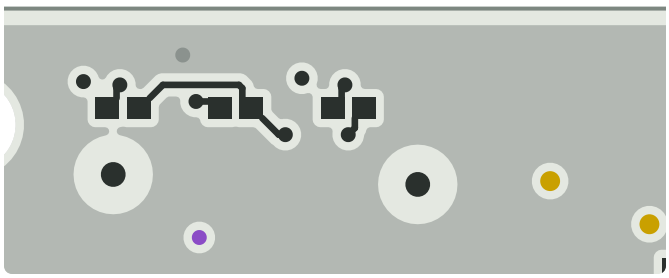
Before, Top



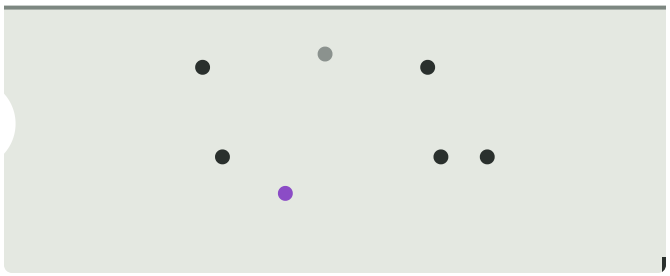
After, Top

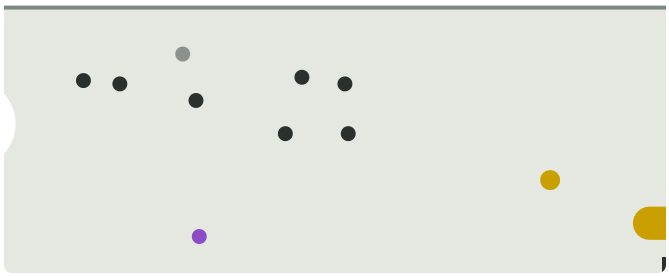


Before, Bottom



After, Bottom



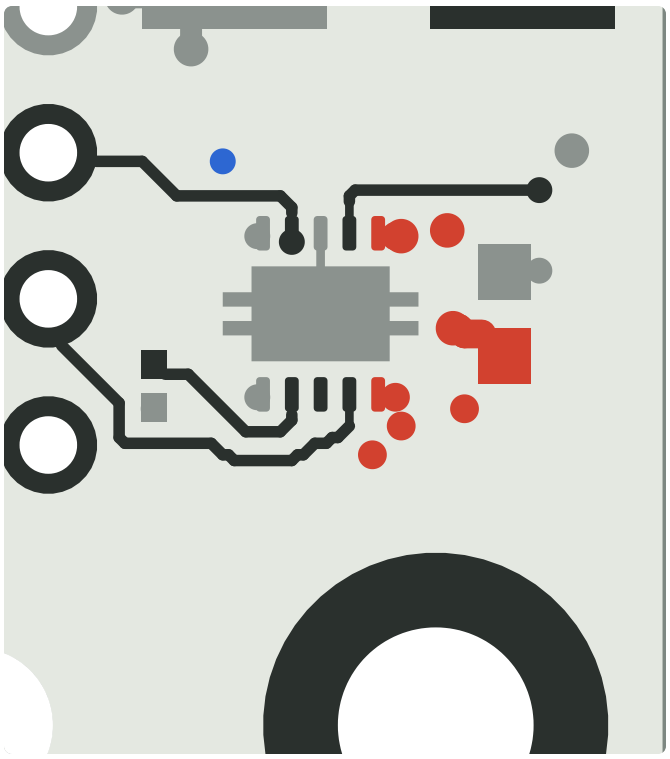


Before, Internal 1

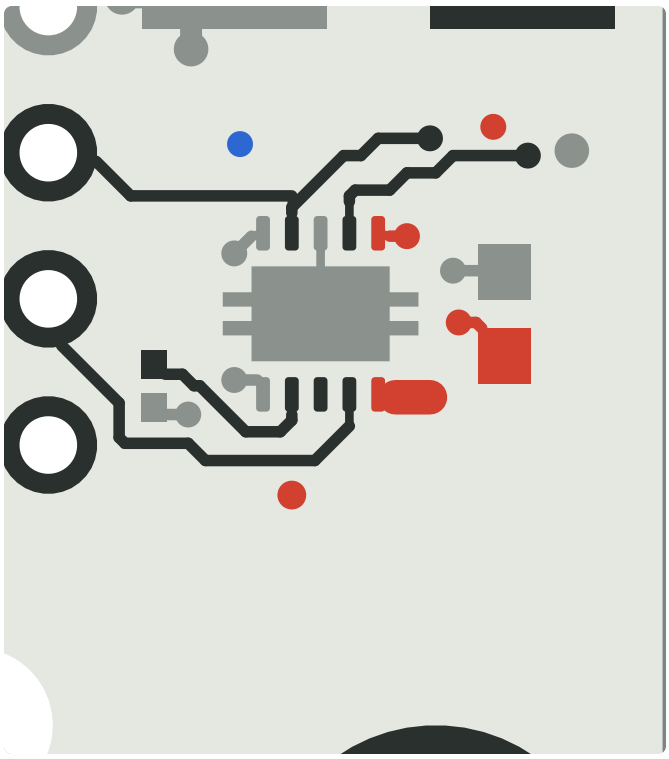
After, Internal 1

u11 payload x 27.0…38.5, y 49.0…62.0 mm

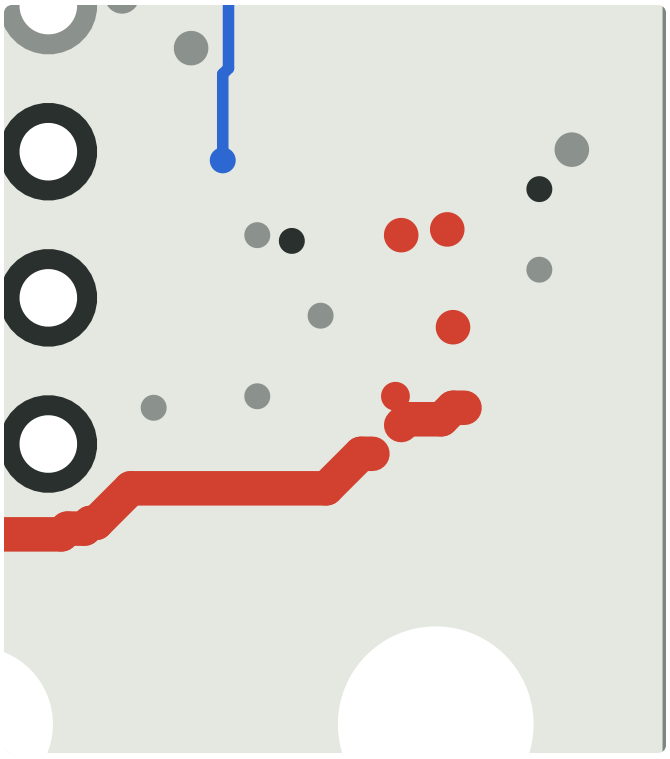
Load switch U11 at the strip's south end: VBUS in (pre-stitched via), VBUS_OUT to J5 (0.6 mm), its signals to J1



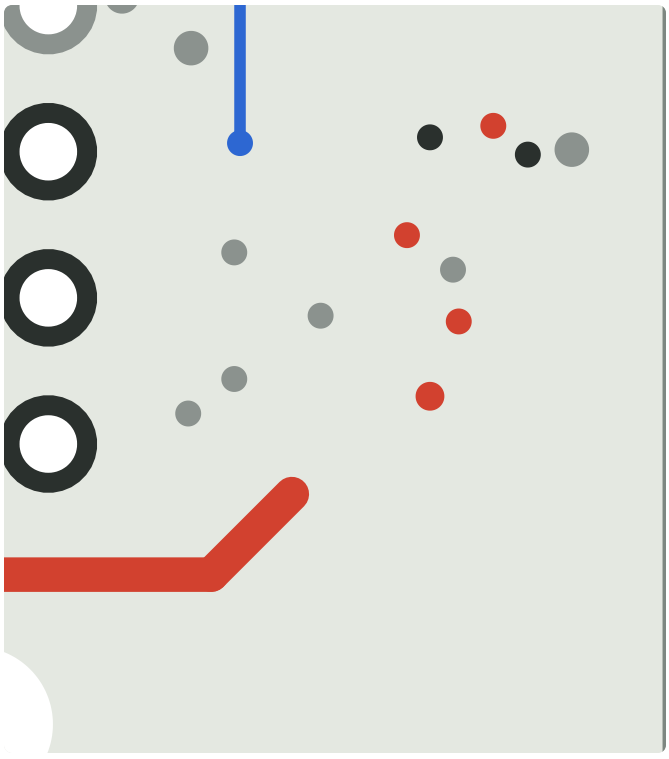
Before, Top



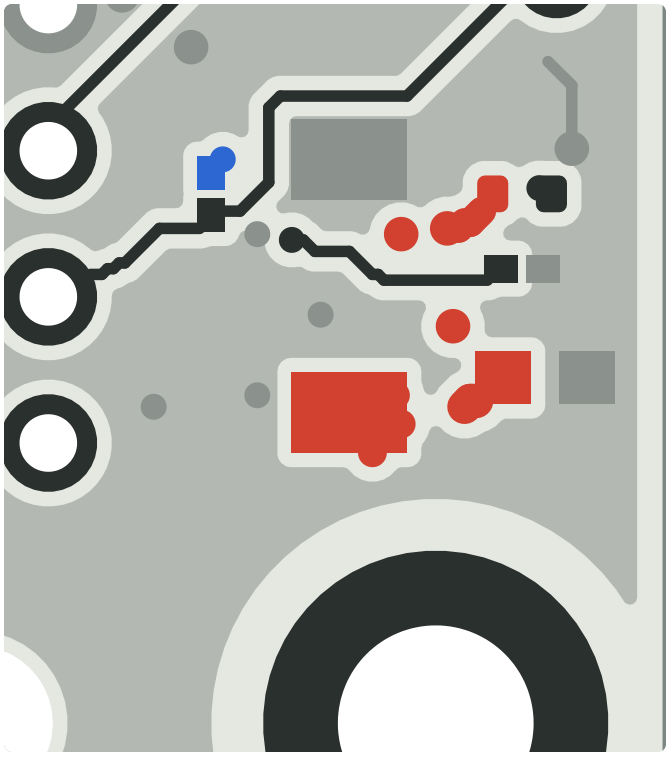
After, Top



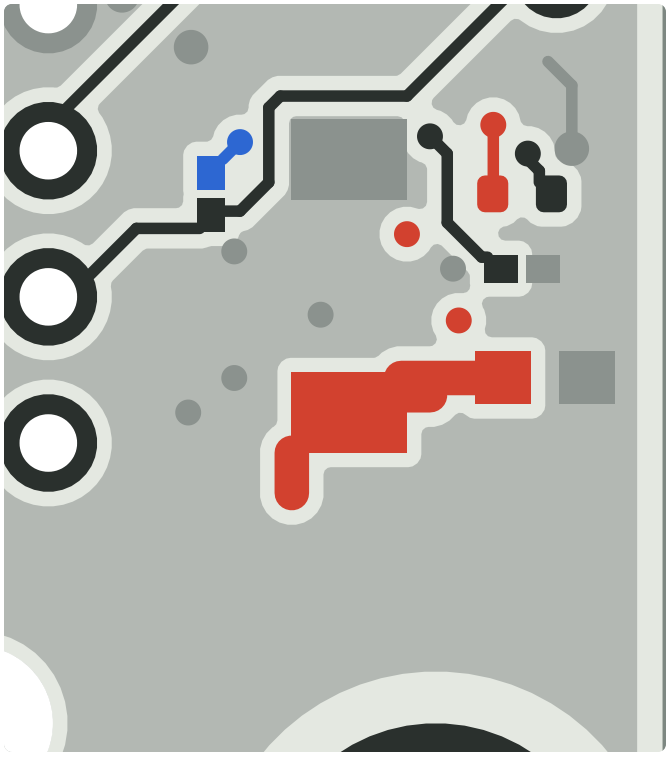
Before, Internal 2 (In3)



After, Internal 2 (In3)



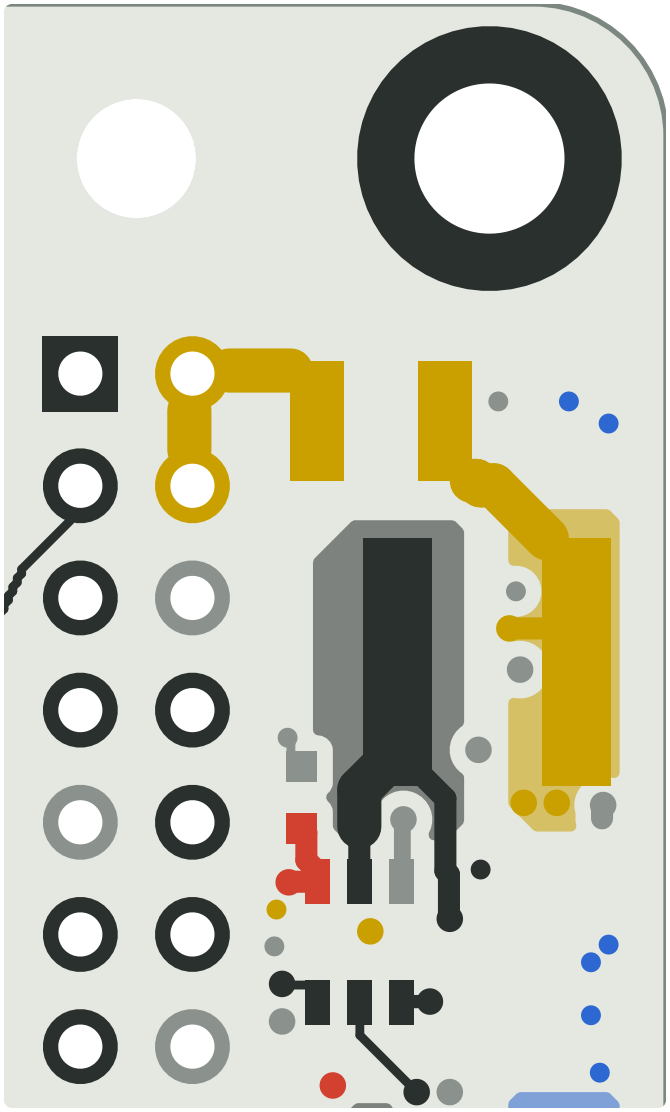
Before, Bottom



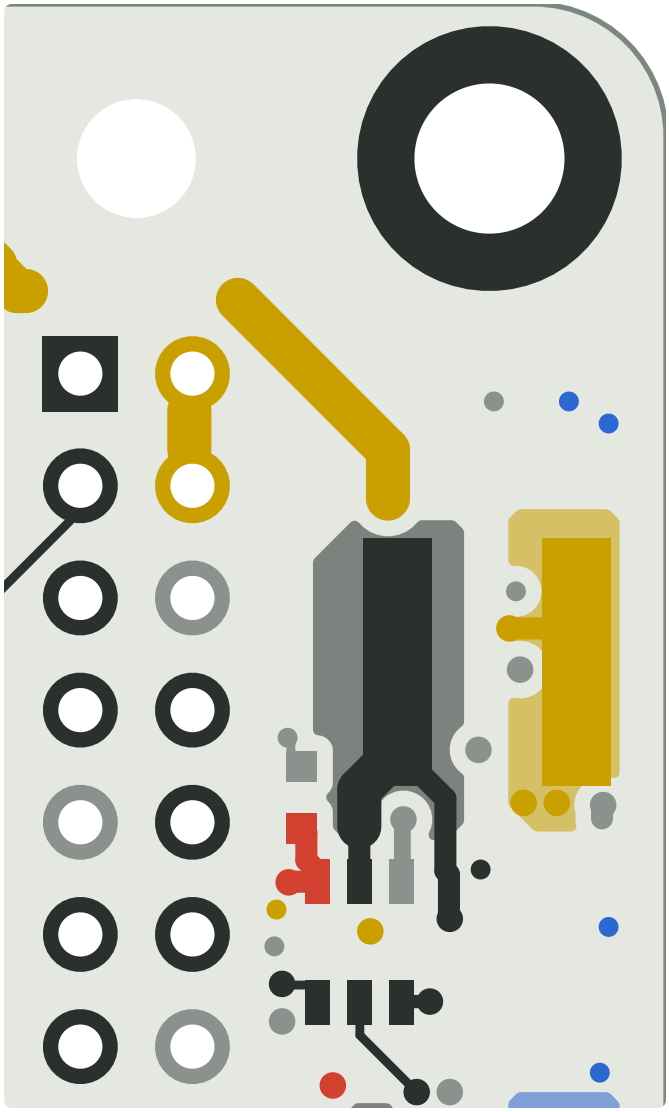
After, Bottom

pi5v strip north x 23.5…38.5, y 0.0…25.0 mm

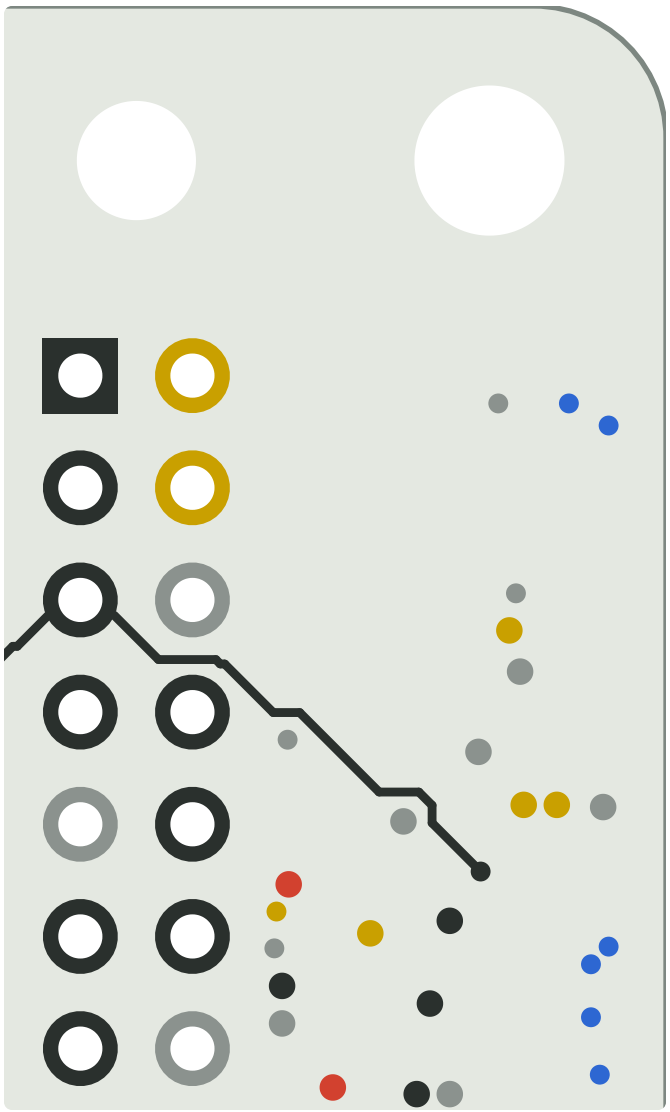
5 V cell at the strip's north end and the 5 V path: L6 -> JP1 in the north band -> J1 pins 2/4 across J1's north end



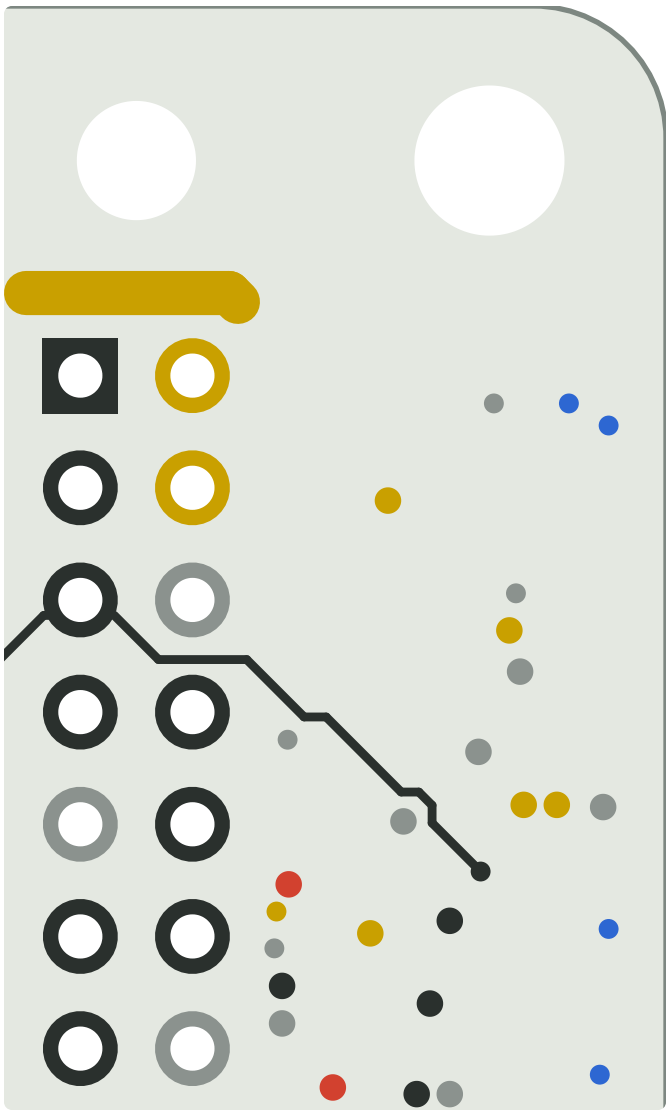
Before, Top



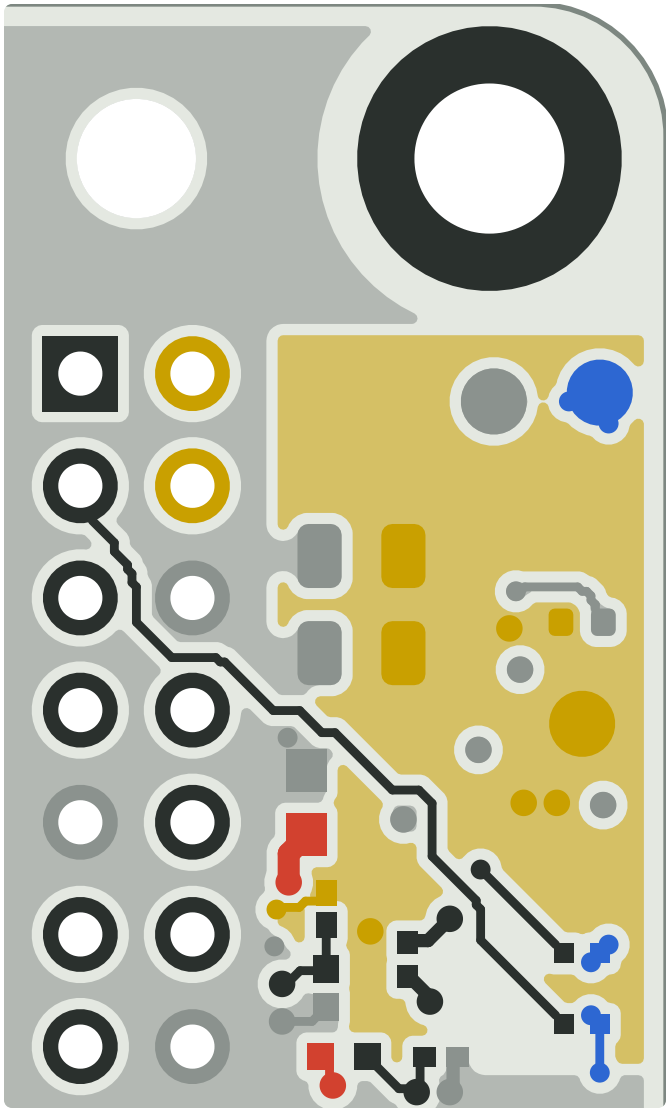
After, Top



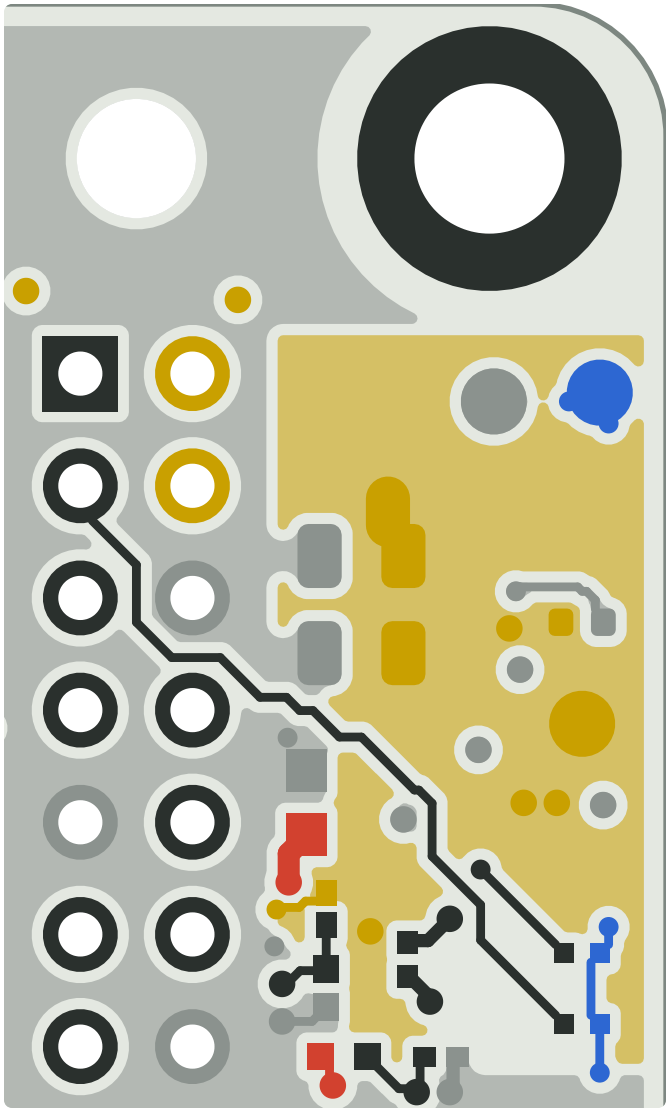
Before, Internal 1



After, Internal 1



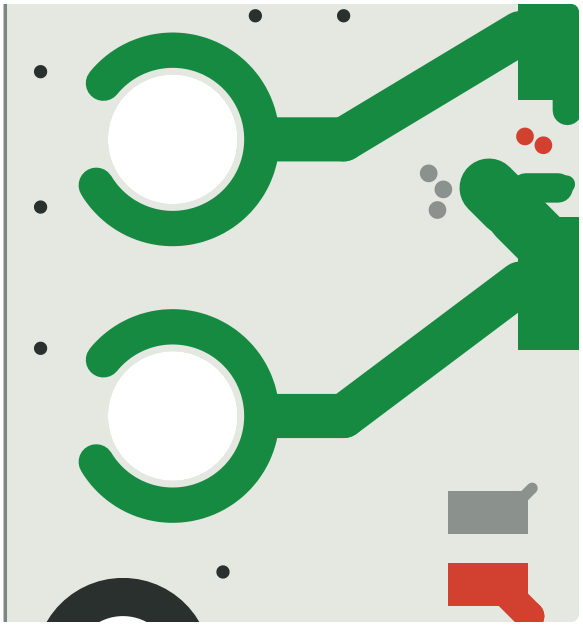
Before, Bottom



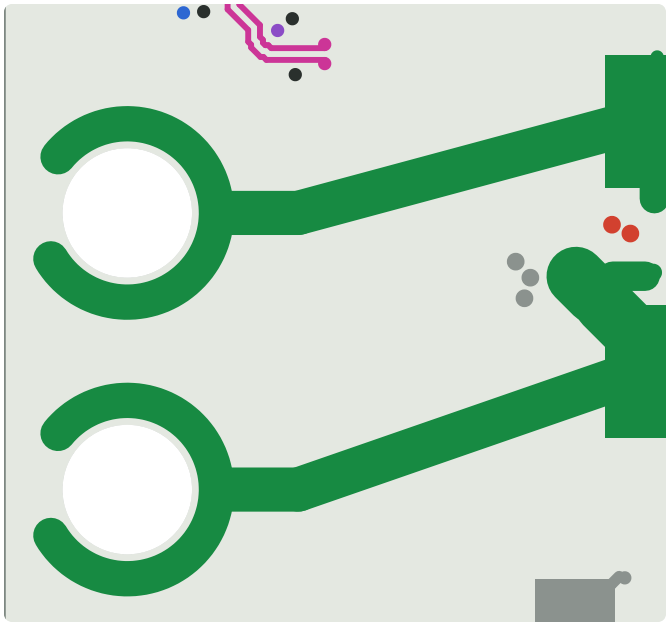
After, Bottom

insert mp1 x -10.5…12.0, y 39.0…60.0 mm

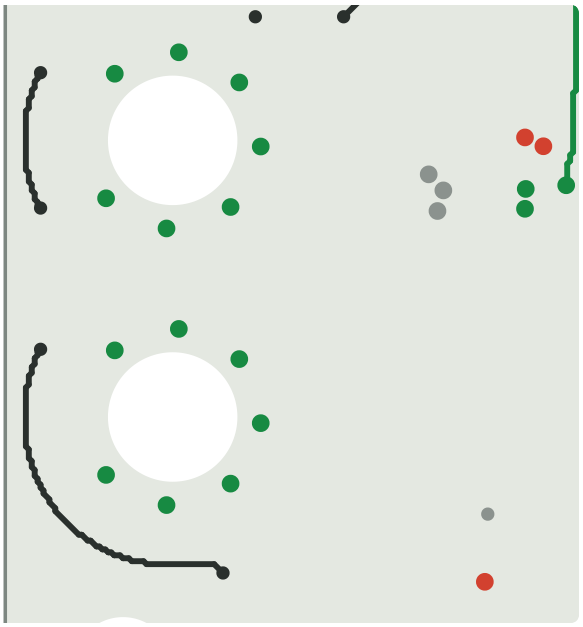
Port-1 inductor L1 (3 mm south) with the insert rings MP1 / MP2 (1.5 mm nearer the wall, 2.5 mm south) and the bus feeds; the P1L block's copper



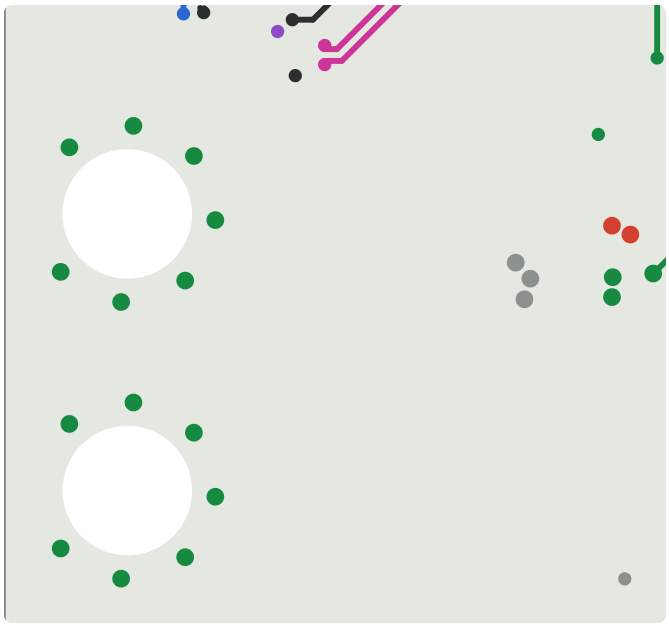
Before, Top



After, Top



Before, Internal 1



After, Internal 1